

# UPES

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|----------------|-------------------------------------------|--------------------|------------------|
| <b>Program</b> | <b>B.Tech. (SOCS/SOAE)</b>                | <b>Semester</b>    | <b>I</b>         |
| <b>Course</b>  | <b>Advanced Engineering Mathematics I</b> | <b>Course Code</b> | <b>MATH 1059</b> |

| UNIT 0: MOTIVATION            |                                                                                                                                                                                                                                                       |                          |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Session 1</b>              | Why study this course? Application of Calculus and Differential equations in addressing real-world problems.                                                                                                                                          |                          |
| UNIT I: DIFFERENTIAL CALCULUS |                                                                                                                                                                                                                                                       |                          |
| <b>Session 2</b>              | Review: Functions and their graphs, polynomial, exponential, and logarithmic functions, Limits, Continuity, and Differentiability                                                                                                                     |                          |
| <b>Session 3</b>              | Rolle's theorem                                                                                                                                                                                                                                       |                          |
| <b>Session 4</b>              | Lagrange's, and Cauchy's mean value theorem                                                                                                                                                                                                           |                          |
| <b>Session 5</b>              | Successive differentiation                                                                                                                                                                                                                            |                          |
| <b>Session 6</b>              | Leibnitz's theorem,                                                                                                                                                                                                                                   |                          |
| <b>Session 7</b>              | Taylor's series without proof                                                                                                                                                                                                                         |                          |
|                               | Lagrange's form of remainder<br><a href="https://www.youtube.com/watch?v=LLHjZ5fhY4U&amp;list=PLbRMhDVUMngeVrxtbBz-n8HvP8KAWBpI5&amp;index=6">https://www.youtube.com/watch?v=LLHjZ5fhY4U&amp;list=PLbRMhDVUMngeVrxtbBz-n8HvP8KAWBpI5&amp;index=6</a> | <b>Flipped Classroom</b> |
| <b>Session 8</b>              | Functions of several variables, Partial differentiation                                                                                                                                                                                               |                          |
| <b>Session 9</b>              | Euler's theorem                                                                                                                                                                                                                                       |                          |
| <b>Session 10</b>             | Jacobian, Maxima and Minima. Recap of Unit I.                                                                                                                                                                                                         |                          |
| UNIT II: INTEGRAL CALCULUS    |                                                                                                                                                                                                                                                       |                          |
| <b>Session 11</b>             | Definite integrals and properties                                                                                                                                                                                                                     |                          |
| <b>Session 12</b>             | Double integrals: Cartesian and Polar coordinates                                                                                                                                                                                                     |                          |
| <b>Session 13</b>             | Change of order of integration                                                                                                                                                                                                                        |                          |
| <b>Session 14</b>             | Change of variables                                                                                                                                                                                                                                   |                          |
| <b>Session 15</b>             | Triple integrals                                                                                                                                                                                                                                      |                          |
| <b>Session 16</b>             | Triple integrals: Cartesian coordinates                                                                                                                                                                                                               |                          |

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| <b>Session 17</b>                            | Triple integrals: Cylindrical coordinates                                                                                                                                                                                                                                  |                          |
|                                              | Triple integrals: Spherical coordinates<br><a href="https://www.youtube.com/watch?v=-I3HUeHi1Ys&amp;list=PLbRMhDVUMngeVrxtbBz-n8HvP8KAWBpI5&amp;index=36">https://www.youtube.com/watch?v=-I3HUeHi1Ys&amp;list=PLbRMhDVUMngeVrxtbBz-n8HvP8KAWBpI5&amp;index=36</a>         | <b>Flipped Classroom</b> |
| <b>Session 18</b>                            | Beta function and their properties                                                                                                                                                                                                                                         |                          |
| <b>Session 19</b>                            | Gamma function and their properties, Applications of Calculus: Real-world problems. Recap of Unit II.                                                                                                                                                                      |                          |
| <b>UNIT III: VECTOR ALGEBRA AND CALCULUS</b> |                                                                                                                                                                                                                                                                            |                          |
| <b>Session 20</b>                            | Motivation, Vector algebra                                                                                                                                                                                                                                                 |                          |
| <b>Session 21</b>                            | Scalar and vector point functions                                                                                                                                                                                                                                          |                          |
| <b>Session 22</b>                            | Vector Differentiation, Gradient                                                                                                                                                                                                                                           |                          |
| <b>Session 23</b>                            | Divergence and Curl                                                                                                                                                                                                                                                        |                          |
| <b>Session 24</b>                            | Vector Integration: Line integral                                                                                                                                                                                                                                          |                          |
| <b>Session 25</b>                            | Vector Integration: Surface integral                                                                                                                                                                                                                                       |                          |
|                                              | Vector Integration: Volume integral<br><a href="https://www.youtube.com/watch?v=EtA0CK8SwkI&amp;list=PLbRMhDVUMngflrZCNOyPZwHUU1pP66vQW&amp;index=60">https://www.youtube.com/watch?v=EtA0CK8SwkI&amp;list=PLbRMhDVUMngflrZCNOyPZwHUU1pP66vQW&amp;index=60</a>             | <b>Flipped Classroom</b> |
| <b>Session 26</b>                            | Application of integrals: Work                                                                                                                                                                                                                                             |                          |
|                                              | Application of integrals: Circulation and Flux<br><a href="https://www.youtube.com/watch?v=kUKpL1YXr0U&amp;list=PLbRMhDVUMngflrZCNOyPZwHUU1pP66vQW&amp;index=56">https://www.youtube.com/watch?v=kUKpL1YXr0U&amp;list=PLbRMhDVUMngflrZCNOyPZwHUU1pP66vQW&amp;index=56</a>  | <b>Flipped Classroom</b> |
| <b>Session 27</b>                            | Green's Theorem and their applications                                                                                                                                                                                                                                     |                          |
| <b>Session 28</b>                            | Stokes theorems and their applications. Recap of Unit-III                                                                                                                                                                                                                  |                          |
|                                              | Gauss divergence theorem and their applications<br><a href="https://www.youtube.com/watch?v=8h3yY0im5XU&amp;list=PLbRMhDVUMngflrZCNOyPZwHUU1pP66vQW&amp;index=59">https://www.youtube.com/watch?v=8h3yY0im5XU&amp;list=PLbRMhDVUMngflrZCNOyPZwHUU1pP66vQW&amp;index=59</a> | <b>Flipped Classroom</b> |

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| Unit IV: Linear Differential Equations with Visualizations |                                                                                                                                                                                                                                                                               |                          |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Session 29</b>                                          | Motivation: Linearity vs non-linearity, Linear superposition principle                                                                                                                                                                                                        |                          |
| <b>Session 30</b>                                          | First-order linear differential equations: Exact differential equations                                                                                                                                                                                                       |                          |
|                                                            | Exact differential equations (Integrating factors)<br><a href="https://www.youtube.com/watch?v=Wx19LXs77mM&amp;list=PLbRMhDVUMngeVrxtbBz-n8HvP8KAWBpI5&amp;index=55">https://www.youtube.com/watch?v=Wx19LXs77mM&amp;list=PLbRMhDVUMngeVrxtbBz-n8HvP8KAWBpI5&amp;index=55</a> | <b>Flipped Classroom</b> |
| <b>Session 31</b>                                          | Applications of first-order linear equations: Radioactive decay, Population growth                                                                                                                                                                                            |                          |
|                                                            | Applications of first-order linear equations: Electric circuits<br><br>Book: G. James, and P. Dyke, "Advanced Modern Engineering Mathematics", 5th Edition, Pearson Education, 2018<br>Page no.: 104                                                                          | <b>Flipped Classroom</b> |
| <b>Session 32</b>                                          | Second-order linear differential equations: Solution of homogeneous equations with constant coefficients                                                                                                                                                                      |                          |
|                                                            | Higher-order linear differential equations: Solution of homogeneous equations with constant coefficients<br><br>Edwards, C. H., & Penney, D. E. (2008). Elementary differential equations. (6th ed.). Pearson.<br>Page no.: 162                                               | <b>Flipped Classroom</b> |
| <b>Session 33</b>                                          | Second and higher order linear differential equations: Solution of non-homogeneous equations with constant coefficients                                                                                                                                                       |                          |
| <b>Session 34</b>                                          | Second and higher order linear differential equations: Solution of non-homogeneous equations with constant coefficients                                                                                                                                                       |                          |
| <b>Session 35</b>                                          | Wronskian                                                                                                                                                                                                                                                                     |                          |
| <b>Session 36</b>                                          | Solution of second-order differential equation by variation of parameters                                                                                                                                                                                                     |                          |
| <b>Session 37</b>                                          | Solution of second-order differential equation by variation of parameters                                                                                                                                                                                                     |                          |
| <b>Session 38</b>                                          | Autonomous vs non-autonomous systems                                                                                                                                                                                                                                          |                          |

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| <b>Session 39</b>                                                    | Applications of Linear ODEs with Visualization Tool:<br><br>GeoGebra<br><br><a href="https://www.geogebra.org/t/differential-equation">https://www.geogebra.org/t/differential-equation</a>         |                          |
| <b>Unit V: Non-Linear Differential Equations with Visualizations</b> |                                                                                                                                                                                                     |                          |
| <b>Session 40</b>                                                    | Brief history of interdisciplinary studies of nonlinear systems                                                                                                                                     |                          |
|                                                                      | Pendulum equations                                                                                                                                                                                  |                          |
| <b>Session 41</b>                                                    | Phase portraits                                                                                                                                                                                     |                          |
| <b>Session 42</b>                                                    | Linearization around equilibrium                                                                                                                                                                    |                          |
|                                                                      | Nonlinear models: Logistic equation<br><br>Edwards, C. H., & Penney, D. E. (2008). Elementary differential equations. (6th ed.). Pearson.<br><br>Page no.: 44                                       | <b>Flipped Classroom</b> |
| <b>Session 43</b>                                                    | Nonlinear models: Bass model of diffusion of innovation with<br><br>GeoGebra<br><br><a href="https://www.geogebra.org/t/differential-equation">https://www.geogebra.org/t/differential-equation</a> |                          |
| <b>Session 44</b>                                                    | Nonlinear models: Lotka-Volterra predator-prey model with GeoGebra<br><br><a href="https://www.geogebra.org/t/differential-equation">https://www.geogebra.org/t/differential-equation</a>           |                          |
| <b>Session 45</b>                                                    | Nonlinear models: Epidemic models-SIR, Application to the nerve impulse.<br><br>Recap of Unit V.                                                                                                    |                          |

### **Disclaimer**

Certain topics in this course are designated as **flipped classroom** to encourage independent learning and deeper understanding. These topics are **mandatory** and will be **assessed** through quizzes, assignments, mid semester and/or end semester examinations. Students are expected to go through the recommended resources and discuss doubts during office hours or tutorial hours of their respective faculties.